Sinusoidal Distribution

Baidurya Bhattacharyya

under the supervision of

Dr. Ayanendranath Basu

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Abstract

Sinusoidal Distribution abstract.

Contents

Contents **1**

1 Introduction **2**

1.1 Objectives 3

1.2 Section Overview and Navigation 3

2 The Sinusoidal Distribution **3**

2.1 Definition 3

2.2 Shape Properties 5

3 Preliminary Mathematical Properties **7**

3.1 Interpretation 7

3.2 Transformation 9

3.3 Joint Density 10

3.4 Normalizing Constant for the symmetric case 12

3.5 CDF for the symmetric case 12

3.6 Concentration Inequalities 13

4 Properties involving the Generalized Hypergeometric Function **13**
4.1 Kernel . . . 14
4.2 Characteristic Function . . . 15
4.3 Moments . . . 17
4.4 CDF and Hazard . . . 18
4.5 Normalizing Constant . . . 18

5 Fitting and Estimation **18**
5.1 MLE, Score equations . . . 19
5.2 MSM on MSM . . . 20
5.3 PDF on PDF . . . 20
5.4 CDF on CDF . . . 20
5.5 Quantile-Density Based Metrics . . . 20

6 Divergence Density/Potential **22**

7 Application **24**
7.1 Analytical approximations . . . 24
7.2 Empirical approximations . . . 24
7.2.1 Proxy distribution for unsimplifiable test-statistic distributions . . . 24
7.2.2 Poorly shaped datasets . . . 24
7.3 Naive or Flexible Bayesian Prior . . . 24
7.4 Flexible Activation Function for NNs . . . 24

A Code **25**
A.1 Choice of Optimizer . . . 25

B Proofs **26**

1 Introduction

Statistical domains have long needed flexible distributions. New candidates that have emerged in this domain include the Generalized Lambda distribution (Chalabi, Scott, and Wuertz 2012),

Most densities in everyday use are unimodal and "bell shaped". The instinctive choice for a bell-shape is the crest of a sine-curve. Even disregarding its elegant mathematical properties, this crest forms a visually natural candidate as a probability density function. Gilbert 1893 mentions the *Sinusoidal distribution* in its most well recognized form to study

the distribution of moon-craters, although Edwards 2000 describes a brief history of the different forms in which Gilbert's proposed density has been written in the past, including in the works of La Grange 1770 and Pearson 1902.

In the context of flexible shape modelling, several attempts have been made to extend the Sine Distribution into a more flexible array of shapes to model a wide variety of data distributions, including Kumar, U. Singh, and S. K. Singh 2015, Mahmood, Chesneau, and Tahir 2019, etc. ¡more to be included based on relevance and priority¿, although such recent modifications utilize mathematically complex extensions of Gilbert's proposed elegant density.

1.1 Objectives

1. Extension of the baseline *Sinusoidal distribution* into a flexible Location-Scale-Skewness-Kurtosis family

2. Provide

1.2 Section Overview and Navigation

1. *Sinusoidal distribution*

2. Mathematical results

3. Fitting examples

4. Other applications

2 The Sinusoidal Distribution

2.1 Definition

Standard Family

Notated

$$Z \sim \text{Sinu}(s, k) \quad Z \in [0, 1], s \geq 0, k \geq 0$$

Kernel

$$\varsigma(z; s, k) = \sin^k(\pi z^s)$$

Normalizing constant

$$A(s,k) = \int_0^1 \sin^k(\pi t^s)\mathrm{d}t$$

Standard PDF

$$\psi(z;s,k) \equiv \frac{\varsigma(z;s,k)}{A(s,k)} = \frac{1}{A(s,k)}\sin^k(\pi z^s)$$

Standard CDF

$$\Psi(z;s,k) = \int_0^z \psi(t;s,k)\mathrm{d}t \equiv \frac{A_z(s,k)}{A(s,k)}$$

Explicit forms of the CDF will be derived in later sections.

Location Scale Extension

Notated $X \sim \mathrm{Sinu}(a,d,s,k) \quad X \in [a, a+d], a \in \mathbb{R}, d \geq 0, s \geq 0, k \geq 0$, X has density:

$$\psi(x;a,d,s,k) = \frac{1}{d}\psi\left(\frac{x-a}{d};s,k\right)$$

Flipped Extension

Notated $X' \sim \mathrm{Sinu}'(a,d,s,k)$, it is defined as $X' :\overset{d}{=} a + d - X$ where $X \sim \mathrm{Sinu}(a,d,s,k)$. Conceivably, it has density

$$\psi'(x;a,d,s,k) = \frac{1}{d}\psi\left(\frac{a+d-x}{d};s,k\right)$$

Multivariate Extension

We provide multivariate extension notions of only the standard *Sinusoidal distribution*, i.e. location scale extensions of the multivariate extensions is left implicit.

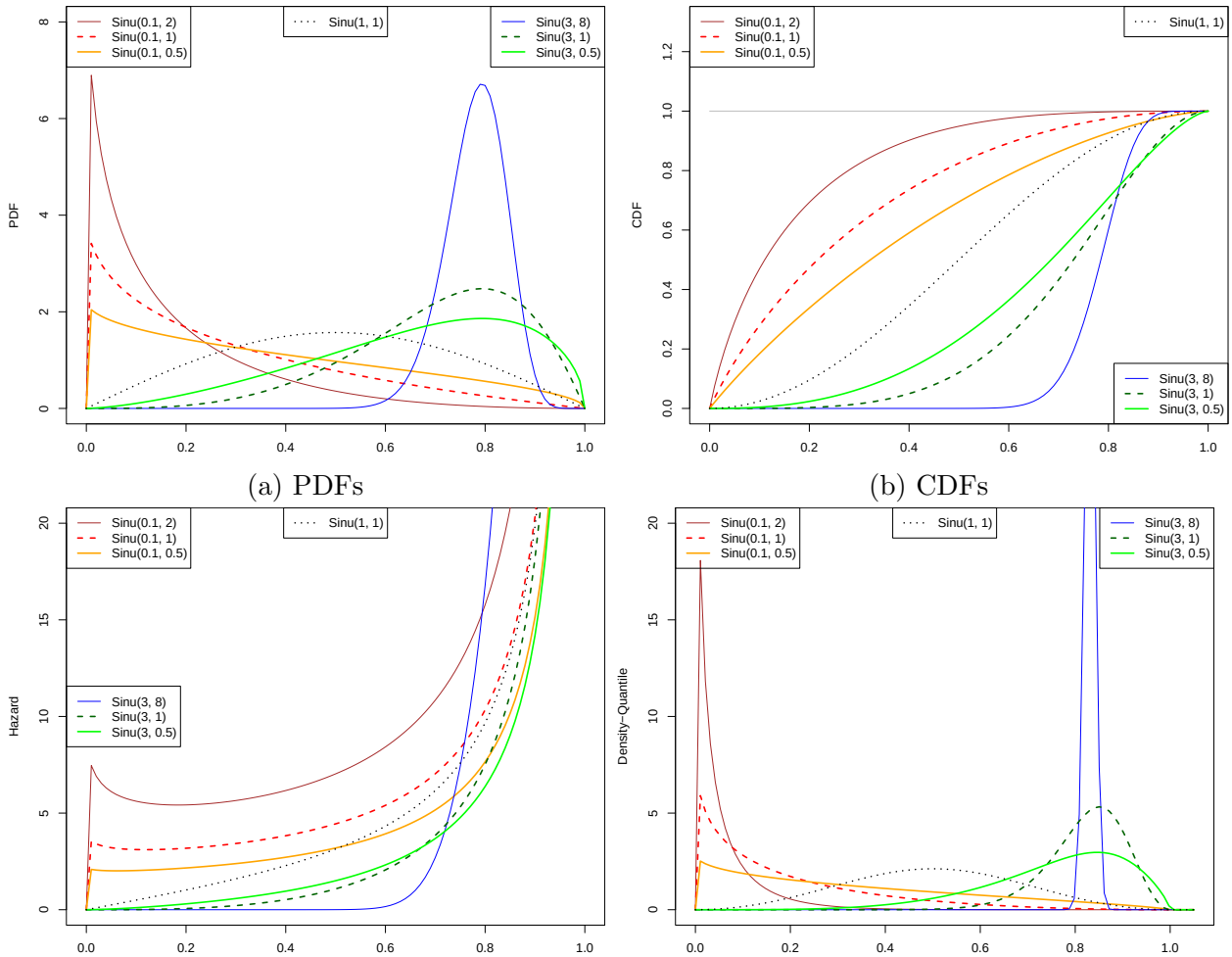
A: Defined independently using individual (s,k)'s: $\vec{Z}_{m\times 1} \sim \psi_m(\vec{s}_{m\times 1}, \vec{k}_{m\times 1})$:

$$\psi_m(\vec{z};\vec{s},\vec{k}) = \prod_{j=1}^{m}\psi(z_j;s_j,k_j)$$

This makes it equivalent to the joint density of m independent Standard Sinusoidal random variables.

B: Defined through a common (s,k) pair via vector generalization

$$\psi_m(\vec{z};\mathbf{D},s,k) = \psi(\vec{z}^\intercal\mathbf{D}^{-1}\vec{z};s,k)$$



(a) PDFs (b) CDFs

(c) Hazard Functions (d) DQFs

Figure 1: Shape plots of different *Sinusoidal distributions*

Unsure of the structure of D, what kind of correlation structure D will provide, whether the defined density will be normalized, how s and k will influence its shape here, etc.

C: Defined through the Archimedean copula.

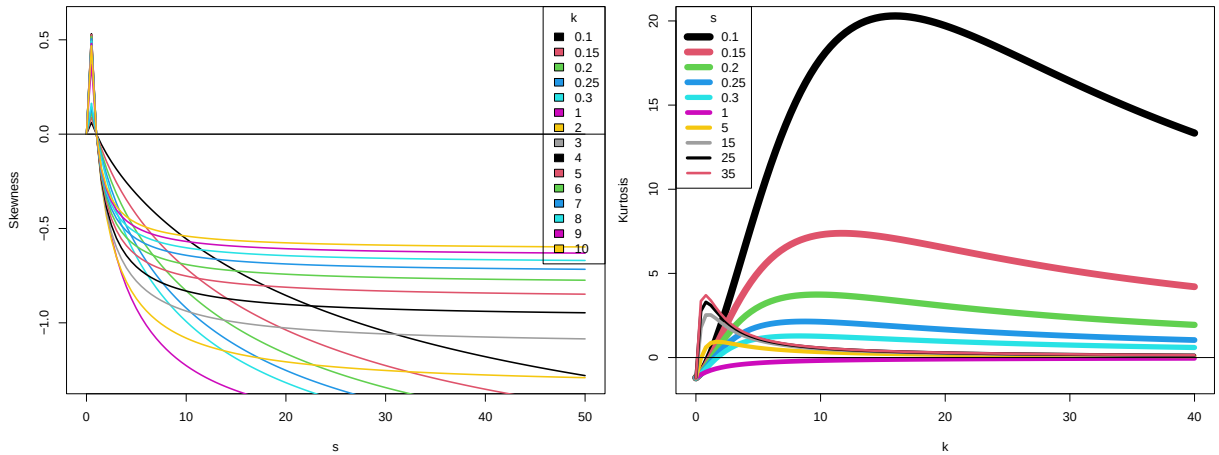
2.2 Shape Properties

Plots of the PDF, CDF, Hazard function and DQF (Density-Quantile Function, as described in Staudte 2017) of some standard *Sinusoidal distributions* are given in Figure 1.

We now describe some shape properties of the Sinu(a, d, s, k) distribution:

1. Support: $[a, a + d]$. a is the location and d is the scale parameter. Alternative parametrization with $b = a + d$ may be considered.

2. s affects skewness of the *Sinusoidal distribution* as follows:



(a) Skewness vs. s

(b) Kurtosis vs. k

Figure 2: s, k jointly affect skewness & kurtosis

(a) $0 \leq s < 1$: left-leaning (positively/right skewed)

(b) $s = 1$: symmetric

(c) $s > 1$: right-leaning (negatively/left skewed)

. The alternative parametrization $s' = \ln s$ calibrates the parameter choice $s = 0$ to the symmetric family, and on either side of it keeps oppositely signed skewness families.

3. The choice $a = -\frac{d}{2}$, $s = 1$ gives the special $\text{Sinu}(-\frac{d}{2}, d, 1, k)$ family, which is symmetric around 0 and supported on the interval $[-\frac{d}{2}, \frac{d}{2}]$.

4. s (or any transformation of it) does not affect the shape of the distribution symmetrically around the point $s = 1$; i.e. $\nexists s' < 1 : \text{Sinu}(s', k) \stackrel{d}{=} 1 - \text{Sinu}(s, k) \quad \forall s > 1$. It is for this reason that we introduced the notion of a flipped *Sinusoidal distribution*.

5. k affects kurtosis non-monotonically.

6. s, k jointly affect both the skewness and the kurtosis. See Figure 2 for reference.

7. Produces a Uniform density when k=0 or s=0

8. Degenerate at a for $d = 0$ (what about $k = \infty$?)

9. $\text{Sinu}(-d/2, d, 1, k(d)) \xrightarrow{d \to \infty} \mathcal{N}(0, 1)$ where $k(d)$ is a certain monotonically increasing function of d (yet to be proven).

10.

Theorem 1. *Mode of* $\text{Sinu}(s, k)$ *is* $2^{-1/s}$ *and the modal density is* $\frac{1}{A(s,k)}$.

Proof.

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}z}\psi(z;s,k) &= \frac{1}{A(s,k)}\frac{\mathrm{d}}{\mathrm{d}z}\sin^k(\pi z^s) \\
&= \frac{1}{A(s,k)}k\sin^{k-1}(\pi z^s)\cos(\pi z^s)\pi s z^{s-1} \\
&= 0 \iff \begin{cases} \sin(\pi z^s) = 0 \implies z = 0, 1 \\ z^{s-1} = 0 \implies z = 0 \\ \cos(\pi z^s) = 0 \implies \pi z^s = \frac{\pi}{2} \implies z = 2^{-1/s} \end{cases}
\end{aligned}$$

And modal density is

$$\psi(2^{-1/s};s,k) = \frac{1}{A(s,k)}\sin^k(\pi z^s)\Big|_{z=2^{-1/s}} = \frac{1}{A(s,k)}$$

□

3 Preliminary Mathematical Properties

Programmatic Notation

We follow these conventions of nomenclature, in both the report and the code as far as possible, to quickly understand the domain of variables, and, in some ways, also their interpretation.

1. $x \in [a, a+d]$
2. $z :\equiv \frac{x-a}{d} \in [0,1]$
3. $u :\equiv z^s \in [0,1]$ also.
4. $\theta :\equiv \pi u \equiv \pi z^s \in [0,\pi]$
5. $\omega := \theta - \frac{\pi}{2} \equiv \pi z^s - \frac{\pi}{2} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$
6. $r = \pm 1$

3.1 Interpretation

We provide a physical interpretation of the *Sinusoidal distribution*; this will be similar to Gilbert 1893, which imagines a circular planet of (constant) radius r in the 2-dimensional

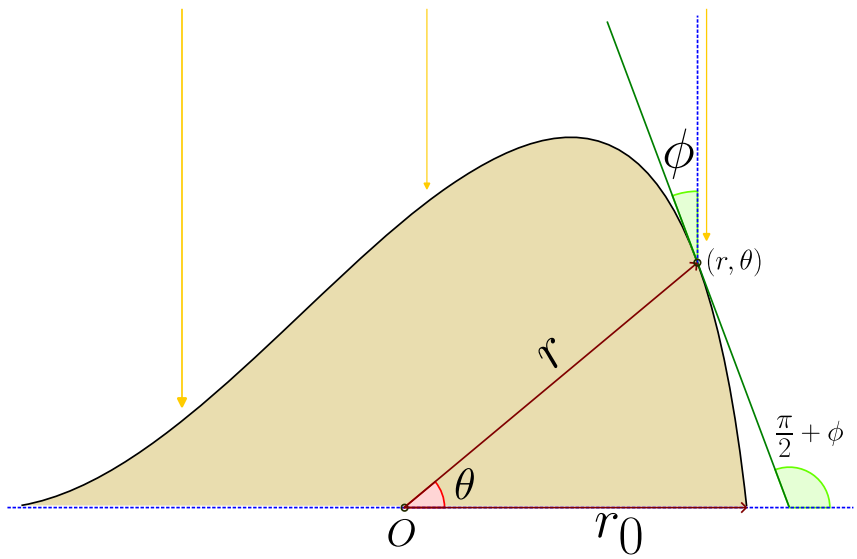


Figure 3: An interpretation of the *Sinusoidal distribution*

plane, and all crater activity happening *vertically* (parallel to Y-axis) onto the planet. Any location (r, θ) on the circumference of the planet that subtends an angle θ at the center, but more importantly also against any *vertical*, receives an intensity (interpreted as proportional to the probability density) of crater activity proportional to $\sin\theta$.

We will show that the *Sinusoidal distribution* $\mathrm{Sinu}(s, k)$ also corresponds to the intensity of crater activity on some non-circular planetary surface, by deriving the functional form of the radial curve $r(z)$ (parametrized by z such that $\theta = \pi z^s$ is the angle every location $r(z)$ on the planet subtends at the origin) which experiences crater activity proportional to $\sin^k(\pi z^s)$ at every $r(z)$.

First, note that both the radius of a planet and the angle subtended at its center are actually immaterial with regard to the intensity of crater activity it experiences when we assume that there is no decay in the intensity of crater activity through space, or atleast in the region of space incident normally upon the planet starting from its north-pole to its equator. In fact, what only matters is the tangential inclination of any location on the planet. To summarise, a location $r(z)$ parametrized as above will experience activity intensity proportional to $\sin(\phi)$, where ϕ is the angle its tangent subtends against the vertical.

Now, consider a planet whose every location $r(z)$ is inclined at an angle $\phi = \sin^{-1}[\sin^k(\pi z^s)]$ against the vertical. it is immediate that the activity intensity there would be $\sin^k(\pi z^s)$.

The characterization of the shape of such a (smooth) radial curve that defines the planet

is a problem in differential geometry. We briefly formulate its solution:

$$x(0) = r_0;\ y(0) = 0$$
$$\phi(z) = \sin^{-1}[\sin^k(\pi z^s)]$$
$$\frac{\mathrm{d}y}{\mathrm{d}x} = \tan\left(\frac{\pi}{2} + \phi\right) \equiv -\cot\phi$$

$$\begin{aligned} x = r(z)\cos(\pi z^s) \quad &\implies \mathrm{d}x = \cos(\pi z^s)\mathrm{d}r - r\sin(\pi z^s)\pi s z^{s-1}\mathrm{d}z \\ y = r(z)\sin(\pi z^s) \quad &\implies \mathrm{d}y = \sin(\pi z^s)\mathrm{d}r + r\cos(\pi z^s)\pi s z^{s-1}\mathrm{d}z \end{aligned}$$

$$\begin{aligned} & \frac{\sin(\pi z^s)\mathrm{d}r + r\cos(\pi z^s)\pi s z^{s-1}\mathrm{d}z}{\cos(\pi z^s)\mathrm{d}r - r\sin(\pi z^s)\pi s z^{s-1}\mathrm{d}z} && = -\cot\left[\sin^{-1}\sin^k(\pi z^s)\right] \\ \implies & \frac{\mathrm{d}r}{r} && = \tan\left[\pi z^s - \sin^{-1}\sin^k(\pi z^s)\right]\cdot \pi s z^{s-1}\mathrm{d}z \\ & && = \tan(\theta - \sin^{-1}\sin^k\theta)\mathrm{d}\theta \\ \implies & \ln r\Big|_{r_0}^{r(z)} && = \int_0^{\pi z^s} \tan(t - \sin^{-1}\sin^k t)\mathrm{d}t \\ \implies & r(z) && = r_0 \exp\left(\int_0^{\pi z^s} \tan(t - \sin^{-1}\sin^k t)\mathrm{d}t\right) \end{aligned}$$

3.2 Transformation

Theorem 2 (Expression as a morphed arcsin beta distribution)**.** *$Y \sim \mathrm{B}(\frac{k+1}{2}, \frac{1}{2})$ and $R \sim \mathrm{Ber}(\frac{1}{2})$, then*

$$Z = \left[R\sin^{-1}\sqrt{Y} + (1-R)(\pi - \sin^{-1}\sqrt{Y})\right]^{\frac{1}{s}}$$

follows a $\mathrm{Sinu}(1,k)$ distribution.

Proof. Consider the transformation $Y = \sin^2(X)$ for $X \in [0, \pi/2]$. The Jacobian of the transformation is:

$$\left|\frac{dx}{dy}\right| = \frac{1}{2\sin(x)\cos(x)} = \frac{1}{2y^{1/2}(1-y)^{1/2}} \tag{1}$$

Substituting the density $f_X(x) \propto \sin^K(x) = y^{K/2}$:

$$f_Y(y) \propto y^{K/2}\cdot y^{-1/2}(1-y)^{-1/2} = y^{\frac{K+1}{2}-1}(1-y)^{\frac{1}{2}-1} \tag{2}$$

This identifies Y as Beta $\left(\frac{K+1}{2}, \frac{1}{2}\right)$. The full support $[0, \pi]$ is obtained by reflecting X across

$\pi/2$ with probability 0.5 via the Bernoulli variable S.

Let $U = \pi Z^S$. The Jacobian of the inverse transformation $z = (u/\pi)^{1/S}$ is $|dz/du| \propto u^{(1-S)/S}$. The density of U is:

$$f_U(u) \propto f_Z((u/\pi)^{1/S}) \cdot u^{(1-S)/S} \propto (u^{(S-1)/S} \sin^K u) \cdot u^{(1-S)/S} = \sin^K u \tag{3}$$

on $u \in [0, \pi]$. By the Beta-transformation theorem for $\sin^K u$, the variable U is generated using $Y \sim \text{Beta}(\frac{K+1}{2}, \frac{1}{2})$. Reversing the power transformation $Z = (U/\pi)^{1/S}$ completes the proof. □

3.3 Joint Density

Theorem 3 (Joint density expression)**.** *For $X_1 \dots X_n \sim \text{Sinu}(a, d, s, k)$, their joint density can be expressed as:*

$$f(x_1 \dots x_n) =$$

Proof.

$$\begin{aligned}
f(x_1 \dots x_n) &= \frac{1}{d^n A^n(s,k)} \left[\prod_{j=1}^{n} \sin \pi \left(\frac{x_j - a}{d} \right)^s \right]^k \\
&= \frac{1}{d^n A^n(s,k)} \left[\prod_{j=1}^{n} \sin \theta_j \right]^k \\
&= \frac{1}{d^n A^n(s,k)} \left[\prod_{j=1}^{n} \cos w_j \right]^k \qquad \text{where } w_j = \pi z_j^s - \frac{\pi}{2} \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]
\end{aligned}$$

Now,

$$
\begin{aligned}
\prod_{j}^{n} \cos w_j &= \prod_{j=1}^{n} \left(\frac{\mathrm{e}^{\mathrm{i}\omega_j} + \mathrm{e}^{-\mathrm{i}\omega_j}}{2} \right) \\
&= \frac{1}{2^n} \sum_{\vec{r} \in \{+1,-1\}^n} \mathrm{e}^{\mathrm{i} \sum_j r_j \omega_j} && \text{[Claim]} \\
&= \frac{1}{2^n} \sum_{(r_1 \ldots r_n) \in \{+1,-1\}^n} \cos \sum_{j=1}^{n} r_j w_j \\
&= \frac{1}{2^n} \sum_{(r_1 \ldots r_n) \in \{+1,-1\}^n} \cos \sum_{j=1}^{n} r_j w_j \\
&= \frac{1}{2^n} \sum_{(r_1 \ldots r_n) \in \{+1,-1\}^n} \cos \left(\pi \sum_{j=1}^{n} r_j z_j^s - \frac{\pi}{2} \sum_{j=1}^{n} r_j \right) && \sum r_j \text{ is the “net valency”} \\
&= \begin{cases} (-1)^{\frac{n}{2}+r} \cos \left(\pi \sum_{j=1}^{n} r_j z_j^s \right) & n \text{ even} \\ (-1)^{\frac{n-1}{2}+r} \sin \left(\pi \sum_{j=1}^{n} r_j z_j^s \right) & n \text{ odd} \end{cases}
\end{aligned}
$$

Alternatively, staying intactly with sin,

$$
\begin{aligned}
\prod_{j=1}^{n} \sin \theta_j &= \prod_{j=1}^{n} \left(\frac{\mathrm{e}^{\mathrm{i}\omega_j} - \mathrm{e}^{-\mathrm{i}\omega_j}}{2} \right) \\
&= \frac{1}{2^n} \sum_{\vec{r} \in \{+1,-1\}^{2n}} \mathrm{e}^{\mathrm{i} \sum_j r_j \omega_j} && \text{[Claim]} \\
&= \frac{1}{2^n} \sum_{(r_1 \ldots r_n) \in \{+1,-1\}^n} \cos \sum_{j=1}^{n} r_j w_j \\
&= \frac{1}{2^n} \sum_{(r_1 \ldots r_n) \in \{+1,-1\}^n} \cos \sum_{j=1}^{n} r_j w_j \\
&= \frac{1}{2^n} \sum_{(r_1 \ldots r_n) \in \{+1,-1\}^n} \cos \left(\pi \sum_{j=1}^{n} r_j z_j^s - \frac{\pi}{2} \sum_{j=1}^{n} r_j \right) && \sum r_j \text{ is the “net valency”} \\
&= \begin{cases} (-1)^{\frac{n}{2}+r} \cos \left(\pi \sum_{j=1}^{n} r_j z_j^s \right) & n \text{ even} \\ (-1)^{\frac{n-1}{2}+r} \sin \left(\pi \sum_{j=1}^{n} r_j z_j^s \right) & n \text{ odd} \end{cases}
\end{aligned}
$$

Proof of proposition by induction

To be done

Then the sum over r_j's partition modularly, so that if $\sum r_j \mod 4$ is 0, then cos, is 1 then sin, is 2 then -cos, is 3 then -sin. However the sum over the first term $r_j z_j^s$ involves convoluted r_j with z_j^s. Thus, no significant simplification may be possible upon this partitioning of the $\sum r_j$ values into modulo 4 groups. In fact this may bring in an excess descriptive specification without adding simplicity to calculation. □

Remark 1 (Pearsonian Random Walk). This is

3.4 Normalizing Constant for the symmetric case

Closed form expression for s=1

Theorem 4 (Calculation of the normalizing constant, symmetric case)**.**

$$A(1,k) \equiv \int_0^1 \sin^k(\pi t)\,\mathrm{d}t = \frac{2^k}{\pi}\mathrm{B}\left(\frac{k+1}{2}, \frac{k+1}{2}\right)$$

Proof.

$$\begin{aligned}\int_0^1 \sin^k(\pi t)\mathrm{d}t &= \int_0^\pi \sin^k\theta\frac{\mathrm{d}\theta}{\pi} \\ &= \frac{1}{\pi}2^k\mathrm{B}\left(\frac{k+1}{2}, \frac{k+1}{2}\right) \qquad \text{[Zwillinger, Eq. 3.621]}\end{aligned}$$

□

3.5 CDF for the symmetric case

Theorem 5 (Indefinite Integral of the Kernel, Symmetric Case)**.** *When $k \in \mathbb{Z}^+$, $s = 1$, we have –*

$$A_z(1,k) = \int_0^z \sin^k \pi t\,dt = \begin{cases} \frac{1}{2^{2n}}\binom{2n}{n}z + \frac{(-1)^n}{\pi 2^{2n-1}}\sum_{k=0}^{n-1}(-1)^k\binom{2n}{k}\frac{\sin 2(n-k)\pi z}{2(n-k)} \\ \frac{1}{\pi 2^{2n}}(-1)^n\sum_{k=0}^{n}(-1)^k\binom{2n+1}{k}\frac{1-\cos(2n+1-2k)\pi z}{2n+1-2k} \end{cases}$$

Proof.

$$\int_0^z \sin^k(\pi t)\mathrm{d}t = \frac{1}{\pi}\int_0^{\pi z} \sin^k\theta \mathrm{d}\theta \quad \left[\theta := \pi t \implies \mathrm{d}t = \frac{\mathrm{d}\theta}{\pi}\right]$$

$$= \frac{1}{\pi}\cdot \left\{\begin{array}{ll} \frac{1}{2^{2n}}\binom{2n}{n}\theta + \frac{(-1)^n}{2^{2n-1}}\sum_{k=0}^{n-1}(-1)^k\binom{2n}{k}\frac{\sin[2(n-k)\theta]}{2(n-k)} & [k=2n] \\ \frac{1}{2^{2n}}(-1)^{n+1}\sum_{k=0}^{n}(-1)^k\binom{2n+1}{k}\frac{\cos(2n+1-2k)\theta}{2n+1-2k} & [k=2n+1] \end{array}\right.\Bigg|_{\theta=0}^{\pi z}$$

[Zwillinger, Eq. 2.513-1, 2]

$$= \frac{1}{\pi}\cdot \left\{\begin{array}{ll} \frac{1}{2^{2n}}\binom{2n}{n}\pi z + \frac{(-1)^n}{2^{2n-1}}\sum_{k=0}^{n-1}(-1)^k\binom{2n}{k}\frac{\sin[2\pi(n-k)z]}{2(n-k)} & [k=2n] \\ \frac{1}{2^{2n}}(-1)^{n}\sum_{k=0}^{n}(-1)^k\binom{2n+1}{k}\frac{1-\cos(2n+1-2k)\pi z}{2n+1-2k} & [k=2n+1] \end{array}\right.$$

□

3.6 Concentration Inequalities

Theorem 6 (*Sinusoidal distribution* is sub-Gaussian)**.** *$X \sim \mathrm{Sinu}(a,d,s,k)$ is sub-Gaussian with parameter $\frac{d^2}{4}$*

Proof. $\because a \le X \le a+d$ a.s., $\therefore \mathbb{E}\left[\mathrm{e}^{\lambda(X-\mathbb{E}X)}\right] \le \exp\left(-\frac{\lambda d^2}{8}\right)$ by Hoeffding's lemma.

Note that because it is sub-Gaussian, it is also sub-Exponential. □

4 Properties involving the Generalized Hypergeometric Function

Introduction to GHF

We will essentially talk about GHgF's (Generalized Hypergeometric Functions), i.e. GHgS's (Generalized Hypergeometric Series's) that converge. We will mention special cases such as Gausssian HgF's and (Kummer's) Confluent HgF's when required.

$$\mathrm{F}_q^p\left(\begin{matrix} a_1 & \cdots & a_p \\ b_1 & \cdots & b_q \end{matrix}; x\right) := \sum_{j=0}^{\infty}\frac{(a_1)^{\overline{j}}\cdots(a_p)^{\overline{j}}}{(b_1)^{\overline{j}}\cdots(b_q)^{\overline{j}}}\frac{x^j}{j!}$$

is defined to be the Generalized Hypergeometric Function calculated at the argument x, where: $(x)^{\overline{n}} := x(x+1)\cdots(x+n-1)$ is the Rising Factorial / Pochhammer function.

Note than if $\mathrm{F}_j :\equiv \frac{(a_1)^{\overline{j}}\cdots(a_p)^{\overline{j}}}{(b_1)^{\overline{j}}\cdots(b_q)^{\overline{j}}}\frac{x^j}{j!}$ denotes the jth term ($j = 0(1)\infty$) of the summation, so that $\mathrm{F}_q^p \equiv \sum_{j=0}^{\infty}\mathrm{F}_j$, then the ratio of consecutive terms of a GHgF defines its parameters

and arguments:

$$\frac{\mathrm{F}_{n+1}}{\mathrm{F}_n}\left[\begin{matrix} a_1 \cdots a_p \\ b_1 \cdots b_1 \end{matrix}; z\right] = \frac{(a_1+n)\cdots(a_p+n)}{(b_1+n)\cdots(b_q+n)}\frac{z}{n+1}$$

We also need the Generalized Binomial expansion formula because k may not be integer. Define from hereon $\binom{r}{k} = \frac{(r)^{\overline{k}}}{k!}$ as the generalized binomial coefficient. Then $(x+y)^r = \sum_{k=0}^{\infty}\binom{r}{k}x^{r-k}y^k$. This series is the same as the finite expansion obtained for $r \in \mathbb{Z}^+$, and is otherwise an infinite series.

4.1 Kernel

Theorem 7 (Kernel as Binomial sum of Euler-form)**.**

$$\sin^k(\pi z^s) = \frac{1}{(2\mathrm{i})^k}\sum_{j=0}^{\infty}\binom{k}{j}(-1)^j \mathrm{e}^{\mathrm{i}(k-2j)\pi z^s} \tag{4}$$

Proof.

$$\begin{aligned}
\sin^k(\pi z^s) &= \sin^k\theta && [\text{let } \theta := \pi z^s] \\
&= \left(\frac{\mathrm{e}^{i\theta} - \mathrm{e}^{-i\theta}}{2i}\right)^k && \because \mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta \\
&= \frac{1}{(2\mathrm{i})^k}\sum_{n=0}^{\infty}\binom{k}{n}\mathrm{e}^{\mathrm{i}\theta(k-n)}(-\mathrm{e})^{-\mathrm{i}\theta n} && \because |\mathrm{e}^{\mathrm{i}\theta}| = |-\mathrm{e}^{-\mathrm{i}\theta}| \text{ and in this case, equality is good enough. W} \\
&= \frac{1}{(2\mathrm{i})^k}\sum_{n=0}^{\infty}\binom{k}{n}(-1)^n\mathrm{e}^{\mathrm{i}\theta(k-2n)} \\
&\equiv \frac{1}{(2\mathrm{i})^k}\sum_{n=0}^{\infty}\binom{k}{n}(-1)^n\mathrm{e}^{\mathrm{i}(k-2n)\pi z^s}
\end{aligned}$$

□

Theorem 8 (Kernel via Generalized Hypergeometric Function)**.**

$$\sin^k(\pi z^s) = (\pi z^s)^k\left({}_0\mathrm{F}_1\left[\begin{matrix} 0 \\ \frac{3}{2} \end{matrix}; -\frac{\pi^2 u^2}{4}\right]\right)^k \tag{5}$$

$$= \mathrm{F}_0^1\left[\begin{matrix} k \\ \cdot \end{matrix}; -\exp(-2\mathrm{i}\pi z^s)\right] \tag{6}$$

Proof. First we work with the Taylor expansion of $\sin\theta$:

$$\begin{aligned}
\sin^k(\pi u) &= (\pi u)^k \left[\sum_{j=0}^{\infty} \frac{(-1)^j (\pi u)^{2j}}{(2j+1)!}\right]^k \\
&= (\pi u)^k \left[\sum_{j=0}^{\infty} \frac{\left(-\frac{\pi^2 u^2}{4}\right)^j}{\left(\frac{3}{2}\right)^{\overline{j}} j!}\right]^k && \because \left(\frac{3}{2}\right)^{\overline{j}} = \frac{(2j+1)!}{2^{2j} j!} \\
&= (\pi u)^k \left({}_0\mathrm{F}_1\left[\begin{matrix}\cdot \\ \frac{3}{2}\end{matrix}; -\frac{\pi^2 u^2}{4}\right]\right)^k
\end{aligned}$$

Alternatively, we construct the required HGF from the ratio of its terms, taking inspiration from section 4. Note that if we were to express the infinite summation in Equation 4 as a HGF, then the ratio of its terms would give:

$$\frac{\mathrm{F}_{n+1}}{\mathrm{F}_n} = (k+n)\frac{-\exp(-2\mathrm{i}\pi z^s)}{n+1}$$

This should be equivalent to the following HGF:

$$\mathrm{F}_0^1\left[\begin{matrix}k \\ \cdot\end{matrix}; -\exp(-2\mathrm{i}\pi z^s)\right]$$

We find that a dual representation is possible, but not much reconciliation or interpretation to this duality is found as of yet. □

4.2 Characteristic Function

Theorem 9. *For $Z \sim \mathrm{Sinu}(s,k)$, its CF is given as –*

$$\varphi_Z(t) = \frac{1}{(2\mathrm{i})^k A(s,k)} \sum_{j=0}^{\infty}\sum_{m=0}^{\infty} \frac{(-1)^j \binom{k}{j} (\mathrm{i}t)^m}{(m+1)!} {}_1\mathrm{F}_1\left[\begin{matrix}\frac{m+1}{s} \\ \frac{m+1}{s}+1\end{matrix}; \mathrm{i}\pi(k-2j)\right] \tag{7}$$

Additionally, the MGF is given as –

$$\phi_Z(t) \equiv \varphi_Z(-it) = \frac{1}{(2\mathrm{i})^k A(s,k)} \sum_{j=0}^{\infty}\sum_{m=0}^{\infty} \frac{(-1)^j \binom{k}{j} t^m}{(m+1)!} {}_1\mathrm{F}_1\left[\begin{matrix}\frac{m+1}{s} \\ \frac{m+1}{s}+1\end{matrix}; \mathrm{i}\pi(k-2j)\right] \tag{8}$$

Proof.

$$\begin{aligned}
\varphi_Z(t) &= \int_0^\infty \mathrm{e}^{\mathrm{i}tz}\psi(z;s,k)\mathrm{d}z \\
&= \frac{1}{A(s,k)}\int_0^1 e^{\mathrm{i}tz}\frac{1}{(2i)^k}\sum_{j=0}^\infty \binom{k}{j}(-1)^j e^{\mathrm{i}\pi(k-2j)z^s}\mathrm{d}z && \text{[Equation 4]} \\
&= \frac{1}{(2i)^k A(s,k)}\sum_{j=0}^\infty (-1)^j\binom{k}{j}\int_0^1 \mathrm{e}^{\mathrm{i}tz}\mathrm{e}^{\pi(k-2j)z^s}\mathrm{d}z && (9)
\end{aligned}$$

Call the inner integral $\mathcal{I}_1 := \int_0^1 \mathrm{e}^{\mathrm{i}tz}\mathrm{e}^{\pi(k-2j)z^s}\mathrm{d}z$. Note that it can be reformulated by expanding $\mathrm{e}^{\mathrm{i}tz}$ in its Taylor expansion, giving –

$$\begin{aligned}
\mathcal{I}_1 &= \int_0^1 \left(\sum_{m=0}^\infty \frac{(\mathrm{i}t)^m z^m}{m!}\right) e^{\mathrm{i}\pi(k-2j)z^s}\mathrm{d}z && \text{Taylor Expansion} \\
&= \sum_{m=0}^\infty \frac{(\mathrm{i}t)^m}{m!}\int_0^1 z^m e^{\mathrm{i}\pi(k-2j)z^s}\mathrm{d}z && \text{Swapping } \sum \text{ and } \int \because \text{ absolute convergence of the Taylor series o} \\
& && (10)
\end{aligned}$$

Call the inner integral $\mathcal{I}_2 := \int_0^1 z^m e^{\mathrm{i}\pi(k-2j)z^s}\mathrm{d}z$ Then, writing $\omega := \pi(k-2j)$ and substituting $u = z^s$, thereby $\implies z = u^{1/s}$ and $\mathrm{d}z = \frac{1}{s}u^{1/s-1}\mathrm{d}u$, we have –

$$\begin{aligned}
\mathcal{I}_2 &= \int_0^1 (u^{1/s})^m e^{\mathrm{i}\omega u}\frac{1}{s}u^{1/s-1}\mathrm{d}u \\
&= \frac{1}{s}\int_0^1 u^{\frac{m+1}{s}-1}e^{i\omega u}\mathrm{d}u \\
&= \frac{1}{m+1}{}_1\mathrm{F}_1\left[\begin{matrix}\frac{m+1}{s}\\ \frac{m+1}{s}+1\end{matrix};\ \mathrm{i}\pi(k-2j)\right] && \text{NIST DLMF, 13.4.E1} && (11)
\end{aligned}$$

So substituting the formulations Equation 10, Equation 11 back into the parent equation Equation 9, we have –

$$\varphi_Z(t) = \frac{1}{(2i)^k A(s,k)}\sum_{j=0}^\infty\sum_{m=0}^\infty \frac{(-1)^j\binom{k}{j}(\mathrm{i}t)^m}{(m+1)!}{}_1\mathrm{F}_1\left[\begin{matrix}\frac{m+1}{s}\\ \frac{m+1}{s}+1\end{matrix};\ \mathrm{i}\pi(k-2j)\right]$$

□

4.3 Moments

Theorem 10.

$$\mu_n(s,k) \equiv E[Z^n] = \frac{1}{(n+1)(2\mathrm{i})^k A(s,k)} \sum_{j=0}^{\infty} (-1)^j \binom{k}{j} \mathrm{F}_1^1 \left[\begin{matrix} \frac{n+1}{s} \\ \frac{n+1}{s}+1 \end{matrix} ; \mathrm{i}\pi(k-2j) \right] \tag{12}$$

Proof. By n^{th} order differentiation of the Characteristic Function Equation 7, since it is known that $E[Z^n] = \frac{1}{\mathrm{i}^n}\left[\frac{\mathrm{d}^n}{\mathrm{d}t^n}\varphi_Z(t)\right]_{t=0}$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\varphi_Z(t)\bigg|_{t=0} = \frac{1}{\mathrm{i}^n} \cdot \frac{1}{(2\mathrm{i})^k A(s,k)} \sum_{j=0}^{\infty} \frac{(-1)^j \binom{k}{j} (\mathrm{i}^n n!)}{(n+1)!} \mathrm{F}_1^1 \left[\begin{matrix} \frac{n+1}{s} \\ \frac{n+1}{s}+1 \end{matrix} ; \mathrm{i}\pi(k-2j) \right]$$

$$\left[\because \frac{\mathrm{d}^n}{\mathrm{d}t^n}(\mathrm{i}t)^m \bigg|_{t=0} = \begin{cases} \mathrm{i}^n \cdot n! & \text{if } m = n \\ 0 & \text{if } m \neq n \end{cases} \right]$$

Alternatively, by direct calculation from PDF:

$$\begin{aligned}
&\int_0^1 z^r \sin^k(\pi z^s)\mathrm{d}z \\
&\quad = \int_0^1 z^r \frac{1}{(2\mathrm{i})^k} \sum_{j=0}^{\infty} \binom{k}{j} (-1)^j \mathrm{e}^{\mathrm{i}(k-2j)\pi z^s} \mathrm{d}z \\
&\quad = \frac{1}{(2\mathrm{i})^k} \sum_{j=0}^{\infty} \binom{k}{j} (-1)^j \int_0^1 \left[\sum_{n=0}^{\infty} z^r \frac{[\mathrm{i}(k-2j)\pi z^s]^n}{n!} \right] \mathrm{d}z \quad \text{[Taylor expansion for e]} \\
&\quad = \frac{1}{(2\mathrm{i})^k} \sum_{j=0}^{\infty} \binom{k}{j} (-1)^j \sum_{n=0}^{\infty} \frac{[\mathrm{i}(k-2j)\pi]^n}{n!(ns+r+1)} \qquad \left[\text{Substitute } \int \text{ and } \sum\right]
\end{aligned}$$

Note that attempts of the following kind do not work elegantly after the 2nd step:

$$\begin{aligned}
E[Z^r] &= \frac{1}{sA(s,k)} \int_0^1 (u^{1/s})^r \sin^k(\pi u) u^{\frac{1}{s}-1} du \\
&= \frac{1}{sA(s,k)} \int_0^1 u^{\frac{r+1}{s}-1} \sin^k(\pi u) du
\end{aligned}$$

□

4.4 CDF and Hazard

Theorem 11.

$$\Psi(z;s,k) = \frac{(\pi z^s)^{k+1}}{\pi s(k+1)A(s,k)}\ \mathrm{F}^1_2\left[\begin{matrix}\frac{k+1}{2} & \\ \frac{3}{2} & \frac{k+3}{2}\end{matrix}; -\frac{\pi^2 z^{2s}}{4}\right] \tag{13}$$

Proof.

$$\Psi(z;s,k) = \frac{1}{A(s,k)}\int_0^z \sin^k(\pi t^s)\mathrm{d}t$$

$$\text{Now } \int_0^z \sin^k(\pi t^s)\mathrm{d}t = \int_0^1 \sin^k(\pi u)\frac{1}{s}u^{1/s-1}\mathrm{d}u \qquad u := t^s \implies t = u^{1/s}; \mathrm{d}t = \frac{1}{s}u^{1/s-1}\mathrm{d}u$$

$$= \frac{1}{s}\int_0^1 \sum ???DEAD\ END$$

□

4.5 Normalizing Constant

Theorem 12 (A(s,k) for s not 1)**.**

$$A(s,k) =$$

Proof.

$$A(s,k) = \int_0^1 \frac{1}{(2\mathrm{i})^k}\sum_{j=0}^{\infty}\binom{k}{j}(-1)^j \mathrm{e}^{\mathrm{i}(k-2j)\pi z^s}\mathrm{d}z \qquad \text{[Equation 4]}$$

$$= \frac{1}{(2\mathrm{i})^k}\sum_{j=0}^{\infty}\binom{k}{j}(-1)^j \int_0^1 \mathrm{e}^{\mathrm{i}(k-2j)\pi z^s}\mathrm{d}z$$

$$=$$

□

5 Fitting and Estimation

This section deals with fitting the *Sinusoidal distribution* over various forms of statistical objects – MSMs (moment summary measures), PDFs, CDFs and ECDFs.

It is to be noted that since the *Sinusoidal distribution* is a location-scale family, investigating its fitting flexibility over standardized/normalized statistical objects is good enough, since it can always adapt to location-scale transformed analogues of the aforesaid objects.

Further, wherever numerical optimization is required, for finiteness of computation, we operate under the following parametric constraints whilst choosing the best candidate *Sinusoidal distribution*, $\text{Sinu}(a, d, s, k)$:

1. $a \in [-10^3, 10^3]$
2. $d \in [10^{-2}, 2 \times 10^3]$
3. $s \in [10^{-2}, 10^2]$
4. $k \in [10^{-2}, 10^2]$

an utilize the `L-BFGS-B` method of the `optim` function found in the `R` programming language.

5.1 MLE, Score equations

$$
\begin{aligned}
l(a, d, s, k; \underset{\sim}{x}) &= \ln \left[\frac{1}{d^n A^n(s,k)} \prod_{i=1}^{n} \sin^k \pi \left(\frac{x_i - a}{d} \right)^s \right] \\
&= -n \ln(d) - n \ln(A(s,k)) + k \sum_{i=1}^{n} \ln \sin \pi \left(\frac{x_i - a}{d} \right)^s
\end{aligned}
$$

Preliminary notation:

$$
\begin{aligned}
A(s,k) &:= \int_0^1 \sin^k(\pi z^s) \, \mathrm{d}z \\
\nabla_s A &= \pi k \int_0^1 \sin^{k-1}(\pi z^s) \cos(\pi z^s)(\ln z) z^s \, \mathrm{d}z \\
\nabla_k A &= \int_0^1 \sin^k(\pi z^s) \ln \sin(\pi z^s) \, \mathrm{d}z
\end{aligned}
$$

The score equations are:

$$
\begin{aligned}
\nabla_a \ell = 0 &\iff \sum_{i=1}^{n} z_i^{s-1} \cot(\pi z_i^s) = 0 \\
\nabla_d \ell = 0 &\iff \pi s k \sum_{i=1}^{n} z_i^s \cot(\pi z_i^s) = -n \\
\nabla_s \ell = 0 &\iff \pi k \sum_{i=1}^{n} \frac{z^s \cos \pi z^s \ln z}{\sin \pi z^s} = n \frac{\nabla_s A}{A} \\
\nabla_k \ell = 0 &\iff \sum_{i=1}^{n} \ln \sin \pi z_i^s = n \frac{\nabla_k A}{A}
\end{aligned}
$$

which may not be solved analytically. We can try figuring out computationally their simultaneous solutions, or we may scrap this entire approach and try maximizing the likelihood computationally through other means.

5.2 MSM on MSM

We tried fitting the Moment Summary Measures (MSMs) of the standard $\text{Sinu}(s, k)$ family over different target MSMs using a specific but arbitrary choice of the loss function: $L(\vec{m}, \vec{m}_t) = (\gamma - \gamma_t)^2 + (\kappa - \kappa_t)^2$. Note that since $\text{Sinu}(a, d, s, k)$ is a location-scale family in (a, d), implying that its first two moments can be freely adjusted by modifying (a, d) alone, it was sufficient to test for the skewness-kurtosis adaptability of the distribution – hence our choice of the standard $\text{Sinu}(s, k)$ family. The results (minimized loss and the choice of parameters) are provided in Table 2.

5.3 PDF on PDF

We tried fitting the PDF of $\text{Sinu}(a, d, s, k)$ over different target PDFs (standard location-scale members only) using numerical optimization and 2 different choices of the loss function (Hellinger distance and L^2 distance). The results (minimized distance with the choice of parameters) are provided in Table 2.

5.4 CDF on CDF

We tried fitting the CDF of $\text{Sinu}(a, d, s, k)$ over different target CDFs (standard location-scale members only) using numerical optimization and 2 different choices of the loss function. The results (minimized distance with the choice of parameters) are provided in Table 3.

5.5 Quantile-Density Based Metrics

Following Staudte 2017, we construct the notion of Divergences based on the density-quantile function $dq(p) = f(F^{-1}(p)) \quad \forall p \in [0, 1]$. The advantages of calculating divergence metrics on this transformed domain is that divergences become location-scale invariant and invariant to "flat" regions in the CDF.

Target Density	Hellinger (a,d,s,k)	L^2 (a,d,s,k)
$\mathcal{N}(0,1)$	0.00004277289 (-20.542932, 34.985498, 1.304535, 100)	
Lognormal$(0,1)$	0.000936801 (0.02827762, 1000, 0.08566970, 35.78972)	
Skew$\mathcal{N}_{\alpha=1}(0,1)$	2.189446e-05 (-7.2216216, 24.8516728, 0.5936823, 100)	
Logistic$(0,1)$	0.003149183 (-80.275639, 102.222513, 2.882431, 100)	
Exp(1)	0.008101119 (0.0005708485, 6.5585684475, 0.0621246472, 1.6766166256)	
DE$(0,1)$	0.01599712 (-8.895071, 16.872288, 1.085768, 15.608238)	
$\Gamma(0.1,1)$	Improper, like below. WHY!	
$\Gamma(100,1)$	WHY! 1 (1.468279e-07, 1.000001e-01, 1.000000e-01, 1.000000e-01)	
B$(1,1)$	0.0030202, (-0.0007780335, 1.0012066335, 0.1004892336, 0.12190192670)	
Cauchy$(0,1)$	0.06878475 (-126.211809, 149.303341, 4.141315, 100)	
$\chi^2(1)$	BAD LIKE GAMMA!	
$\chi^2(10)$	5.791378e-06 (-0.3201950, 97.1337699, 0.2815837, 22.0271292)	
$\mathrm{t}_1(0,1)$	0.06878477 (-126.114092, 149.207078, 4.138119, 100)	
$\mathrm{t}_{10}(0,1)$	INTEGRAL DIVERGENT	
F$(1,1)$	BAD OUTPUT	

Table 1: Some common target densities and the best candidate *Sinusoidal distribution*

Target Density	Hellinger (a,d,s,k)	L^2 (a,d,s,k)
$\mathcal{N}(0,1)$	0.00004277289 (-20.542932, 34.985498, 1.304535, 100)	
Lognormal$(0,1)$	0.000936801 (0.02827762, 1000, 0.08566970, 35.78972)	
Skew$\mathcal{N}_{\alpha=1}(0,1)$	2.189446e-05 (-7.2216216, 24.8516728, 0.5936823, 100)	
Logistic$(0,1)$	0.003149183 (-80.275639, 102.222513, 2.882431, 100)	
Exp(1)	0.008101119 (0.0005708485, 6.5585684475, 0.0621246472, 1.6766166256)	
DE$(0,1)$	0.01599712 (-8.895071, 16.872288, 1.085768, 15.608238)	
$\Gamma(0.1,1)$	Improper, like below. WHY!	
$\Gamma(100,1)$	WHY! 1 (1.468279e-07, 1.000001e-01, 1.000000e-01, 1.000000e-01)	
B$(1,1)$	0.0030202, (-0.0007780335, 1.0012066335, 0.1004892336, 0.12190192670)	
Cauchy$(0,1)$	0.06878475 (-126.211809, 149.303341, 4.141315, 100)	
$\chi^2(1)$	BAD LIKE GAMMA!	
$\chi^2(10)$	5.791378e-06 (-0.3201950, 97.1337699, 0.2815837, 22.0271292)	
$\mathrm{t}_1(0,1)$	0.06878477 (-126.114092, 149.207078, 4.138119, 100)	
$\mathrm{t}_{10}(0,1)$	INTEGRAL DIVERGENT	
F$(1,1)$	BAD OUTPUT	

Table 2: Some common target densities and the best candidate *Sinusoidal distribution*

6 Divergence Density/Potential

For some target density g, if $D_g(f) = \int_{x\in\mathcal{X}} L(f(x), g(x))\mathrm{d}x$ denotes the divergence of candidate density f from target g, then define:

$$V(x,y) = L(y, g(x))$$

Target CDF	Hellinger (a,d,s,k)	L^2 (a,d,s,k)
$\text{Bin}(10, 0.1)$		
$\text{Bin}(10, 0.9)$		
$\text{Poi}(1)$		
$\text{Poi}(5)$		
$\text{Geo}(0.1)$		
$\text{Geo}(0.9)$		
$\text{NB}(10, 0.1)$		
$\text{NB}(10, 0.9)$		
$\text{Hyp}(20, 16, 3)$		
$\text{Hyp}(20, 6, 4)$		

Table 3: Some common target CDFs and the best candidate *Sinusoidal distribution*

as the divergence density/potential associated with g and induced by L. It gives a glimpse at the topography of penalization of $D(\cdot, \cdot)$ with respect to g. This is a useful metric if we want to compare how different tuning parameter α's affect the DPD. The process of fitting

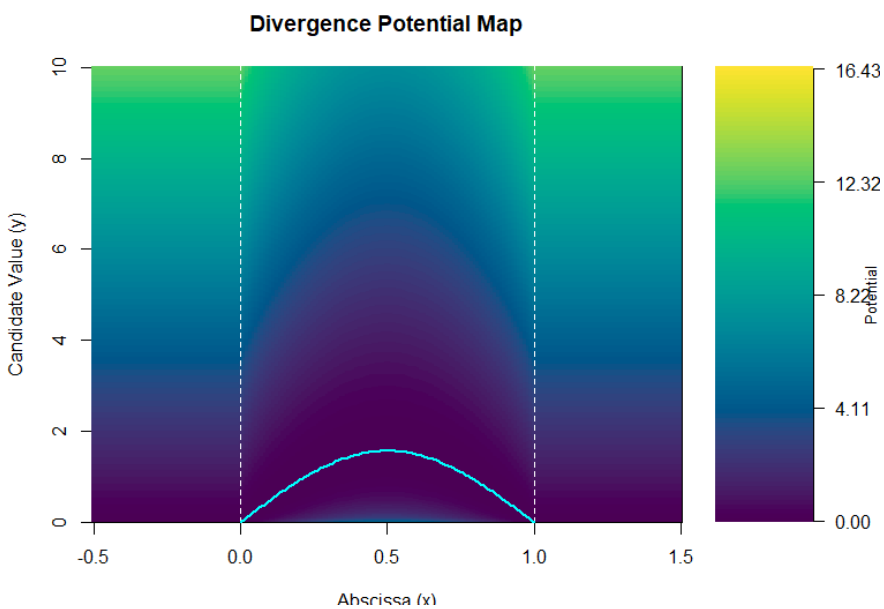


Figure 4: Divergence Potential associated with $\text{Sinu}(1, 1)$ and induced by the Hellinger distance

a candidate distribution over a given target then becomes merely an act of minimizing the line integral of potential function along the density line whilst ensuring that the area under the density line is 1.

We can conclude that the *Sinusoidal distribution* is indeed more flexible if for different

target densities, which may even come from data, it can minimize the divergence better than conventional distributions. At the same time, excess efficiency implies lack of robustness in the presence of contamination. In such cases, we justify the utility of the *Sinusoidal distribution* by saying that it fits to data more flexibly than other distributions, and transparently displays the nature of the underlying divergence metric without interference from the rigidity of its shape.

An apparent benefit of the Divergence Potential over the standard choice of Influence Function in Robust Statistics literature is that it provides more information about the divergence functional than possible through incorporation of a degenerate contamination and subsequent analysis of the limit for an infinitesimal perturbation of that contamination.

7 Application

7.1 Analytical approximations

7.2 Empirical approximations

7.2.1 Proxy distribution for unsimplifiable test-statistic distributions

7.2.2 Poorly shaped datasets

7.3 Naive or Flexible Bayesian Prior

7.4 Flexible Activation Function for NNs

Check if sinu CDF activation function can match a tanh/sigmoid activation function, and how good in the 0,1 range. if support is increased then does it do better?

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A Code

Code for this project is available in github.com/BoiBhai/Sinusoidal-Distribution.git.

A.1 Choice of Optimizer

Following optimization techniques were tried and their drawbacks realized:

1. `optim`'s `L-BFGS-B` method: sensitivity on initial choices of parameters.

2. `optim`'s `Nelder-Mead` method: unavailability of bounding boxes and the presence of unbounded solutions for certain problems such as $\mathcal{N}(0, 1)$ fitting.

B Proofs